

RECTUM

The upper part of the rectum is **intrapertitoneal**, then the midpart is **semi-intrapertitoneal** and the lowest part is **infrapertitoneal**

*Divided in 3 parts according to the blood supply
(NOT THE SAME DIVISIONING AS ABOVE)*

NO taeniae

Sacral and perineal flexures

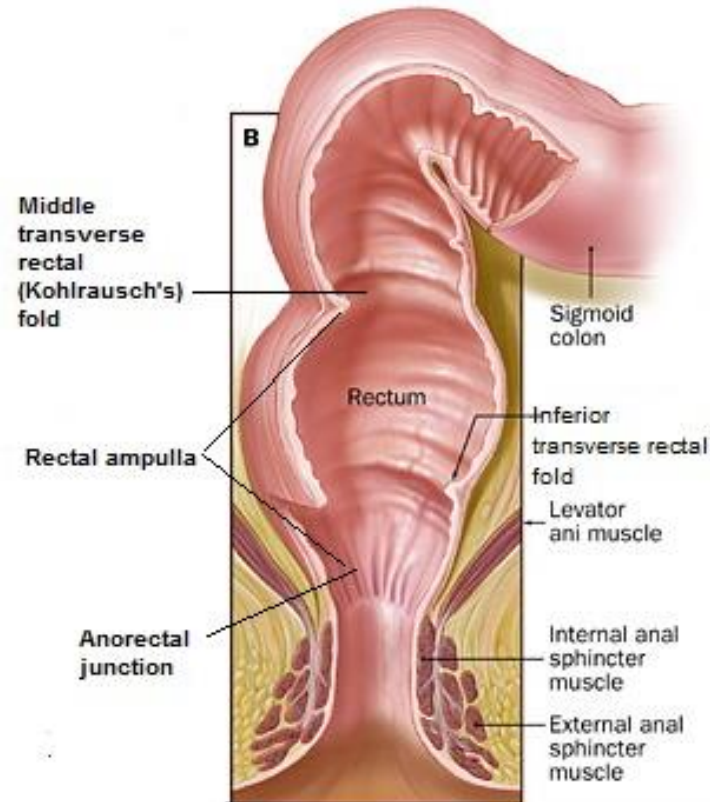
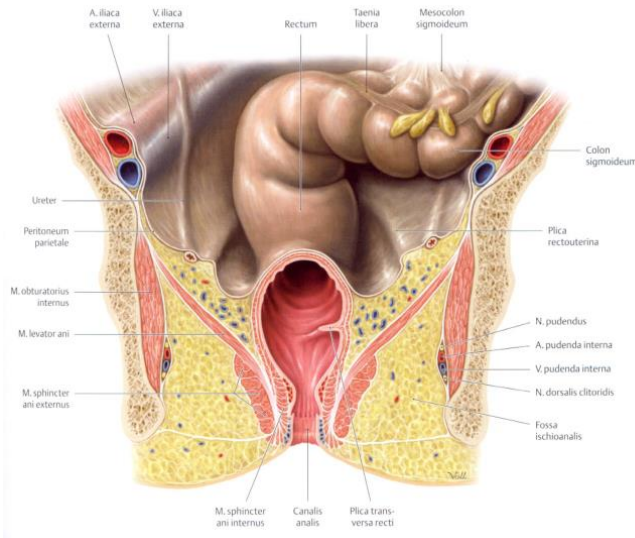
Rectal ampulla between the **middle and inferior transverse folds**

Connected to the sacrum by connective tissue

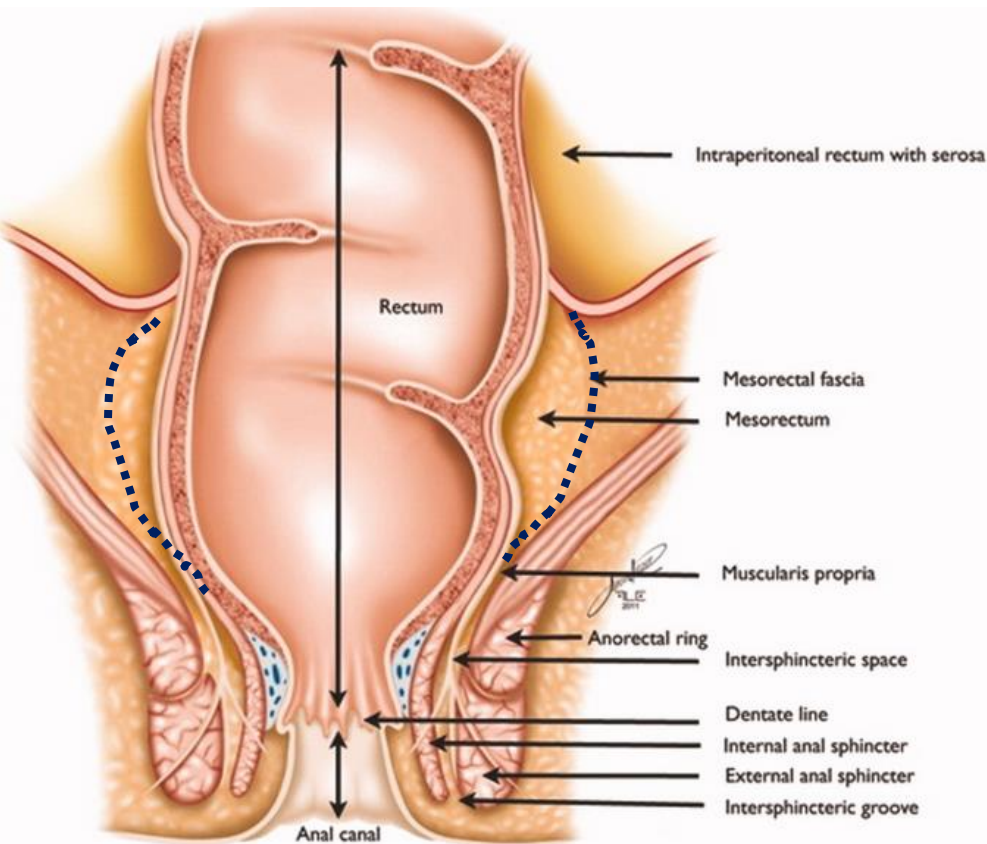
Touches the ileum, sigmoid colon; the perineal flexure in **M**: touches the bladder and the seminal vesicle, fuses with the prostate and in **F**: touches the posterior fornix of vagina

Lowest part sits into the levator ani (pelvic floor)

*Develops from the hindgut (endoderm) above
and from ectoderm below the anorectal junction*



RECTUM & ANAL CANAL



The **mesorectum** (mesentery of the rectum) stretches along the rectum and is enclosed by the mesorectal fascia.

It is a **fatty lymphovascular tissue**

Content: *superior rectal vessels and their branches, lymphatic vessels and nodes, inferior mesenteric plexus and loose connective tissue.*

ANAL CANAL

lies below the ampulla, approx 4-5 cm corresponds to the perineal flexure
Completely closed (except for defecation)
Divided by the pectinate line (pecten)

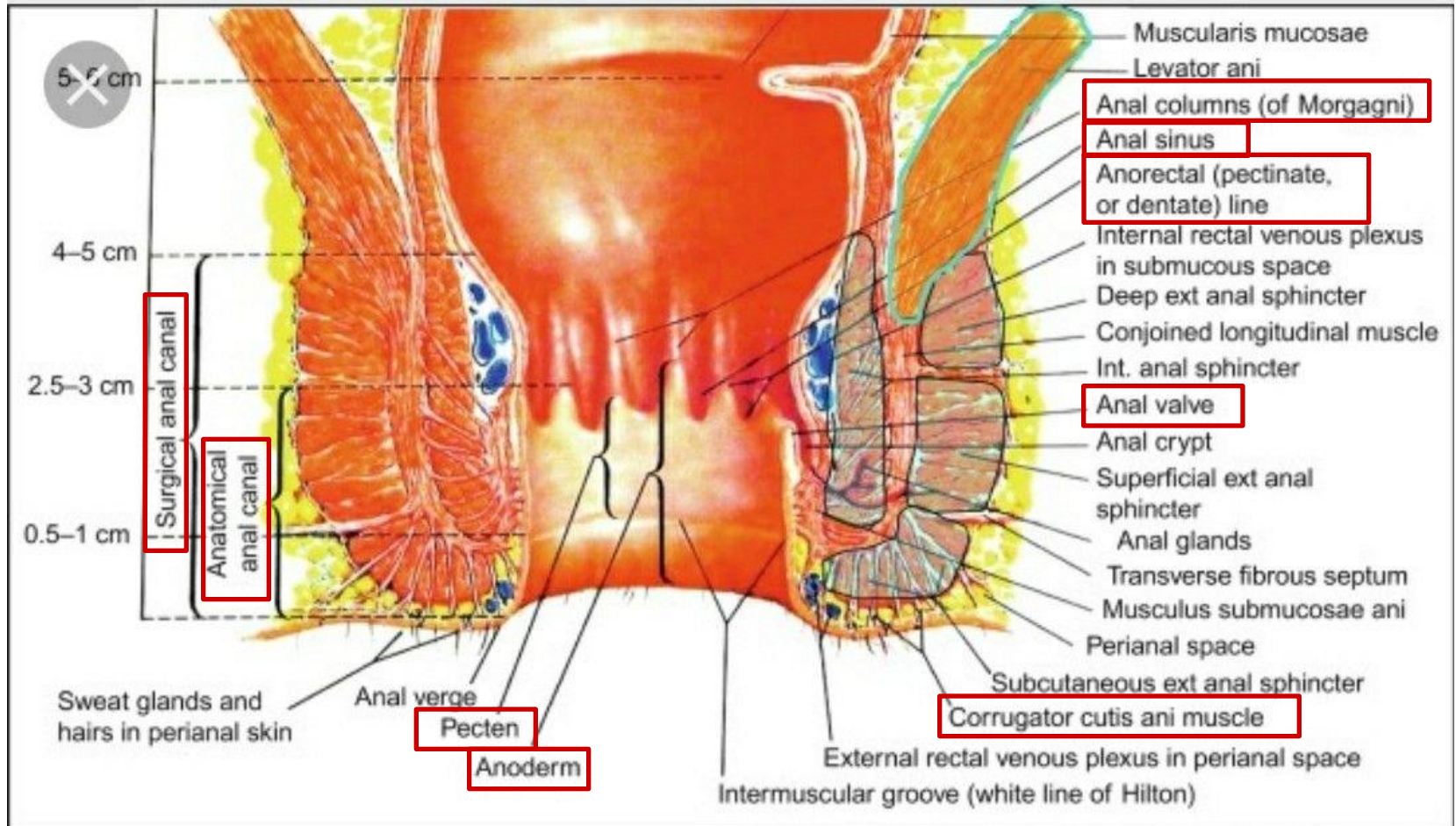
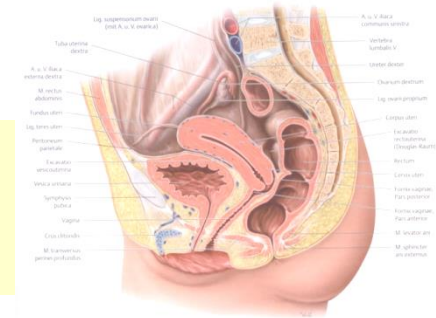
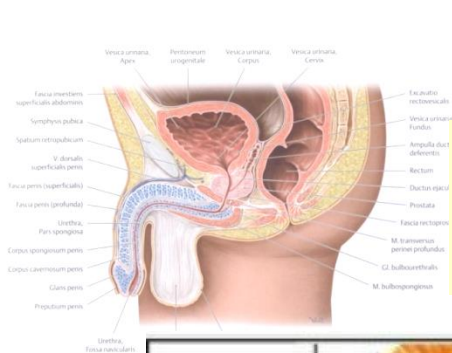
-- upper part - **ENDODERMAL**

columns of Morgagni, haemorrhoidal vv. , columnar epithelium, autonomic innervation - no sensory perception), drains to the paraaortic lymph nodes, **adenocarcinoma**

-- lower part – **ECTODERMAL**, squamous epithelium, somatic innervation – sensitive to touch, drains to the superficial inguinal lymph nodes, squamous epithelial carcinoma

RECTUM & ANAL CANAL

Why is the rectal digital examination important ? What is it?



BLOOD SUPPLY OF THE RECTUM

ARTERIES

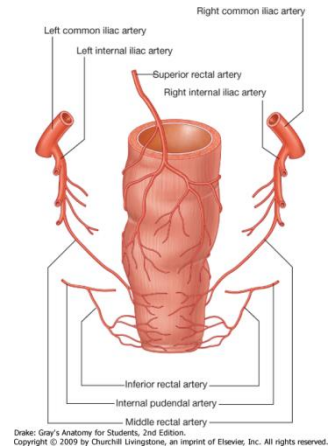
Superior rectal (INF. MESENTERIC A.)
Supplies the ampulla

Middle rectal (INTERNAL ILIAC A.)
Supplies the lower intrapelvic part and the levator ani

Inferior rectal (INTERNAL PUDENDAL A.)
Supplies the extrapelvic part – ext anal sphincter

+ **median sacral a.**

The branches form anastomoses with each other



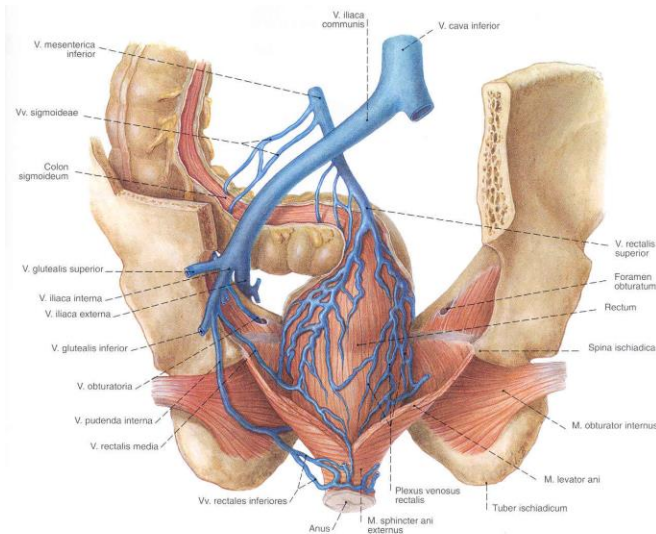
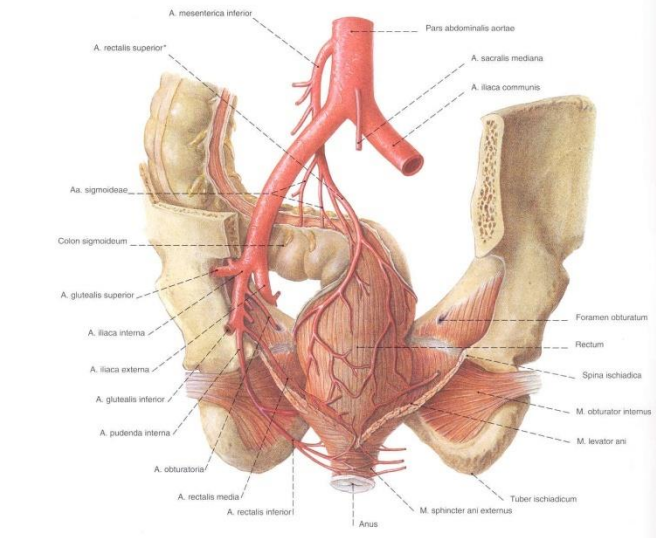
VEINS

Superior rectal v. - unifies from 3 tributaries – **UNPAIRED**,
drains to the inferior mesenteric vein - **PORTAL V.**

Middle rectal v. - **PAIRED** – drains to the internal iliac v -
INFERIOR V. CAVA

Inferior rectal v. - **PAIRED** - leads to the internal pudendal v -
INFERIOR V. CAVA)

BYPASSING THE LIVER - QUICK CENTRAL EFFECT
- use of suppositories

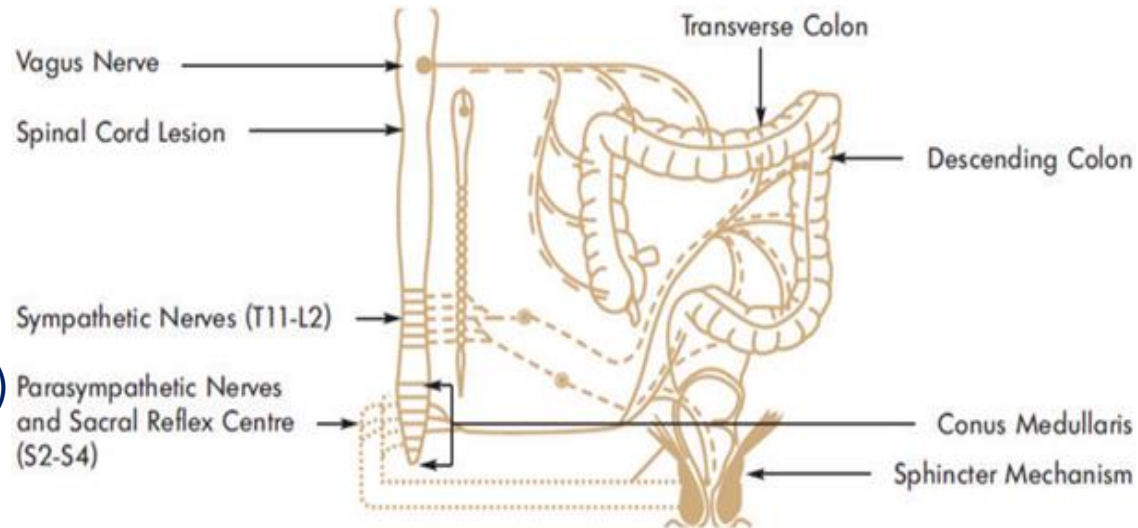


INNERVATION

PARASYMPATHETIC INNERVATION
(secretomotor and motor the to colic muscles)

Vagus nerve innervates the colon to the splenic flexure (Cannon-Böhm's point).

Sacral parasympathetic fibres (S3-S5) innervate the descending colon, sigmoid colon and rectum.



SYMPATHETIC INNERVATION

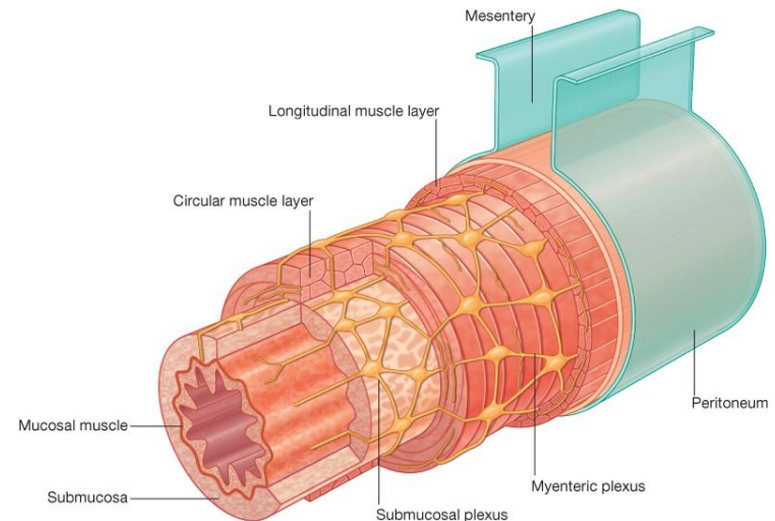
(vasoconstrictor and inhibitory to the colic muscles)

superior and inferior mesenteric plexus

greater and lesser splanchnic nerves (T6-11 ganglia)

Enteric nervous system:

Meissner's plexus and Auerbach's plexus



Drake: Gray's Anatomy for Students, 2nd Edition.
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Nodus hemorrhoidalis



Inflamed hemorrhoids

Hemorrhoids are enlarged veins located within tissues of the lower portion of the rectum or anus

ADAM.

Internal hemorrhoid



Origin above dentate line
(internal rectal plexus)

External hemorrhoid

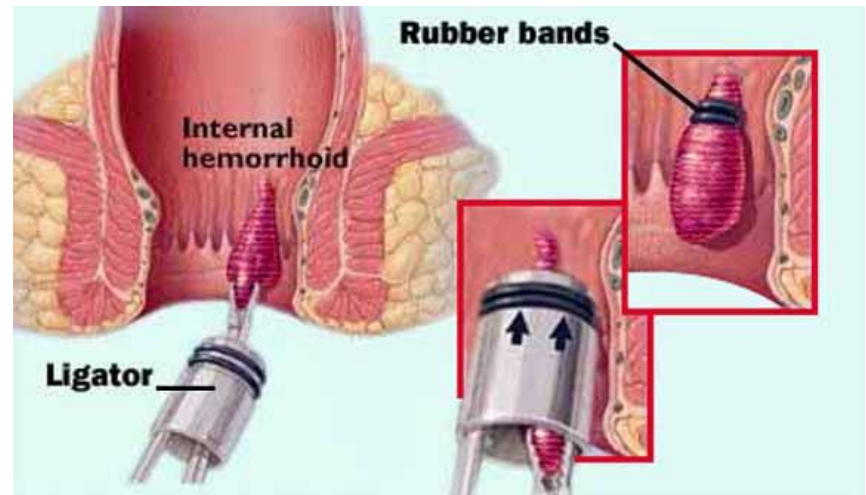


Origin below dentate line
(external rectal plexus)

Mixed hemorrhoid



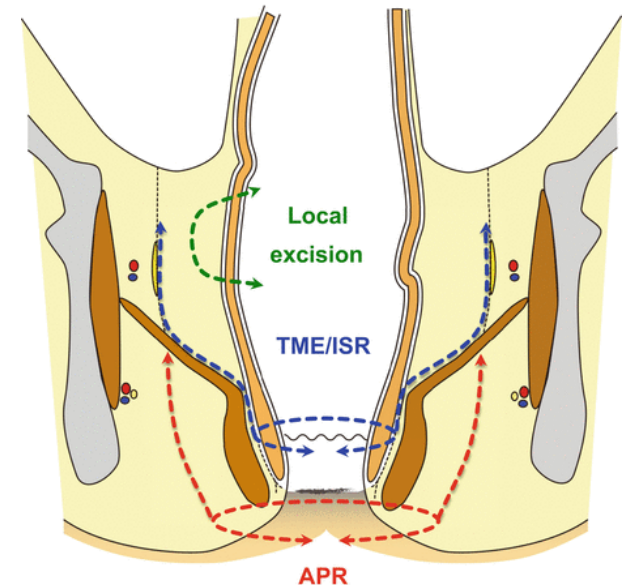
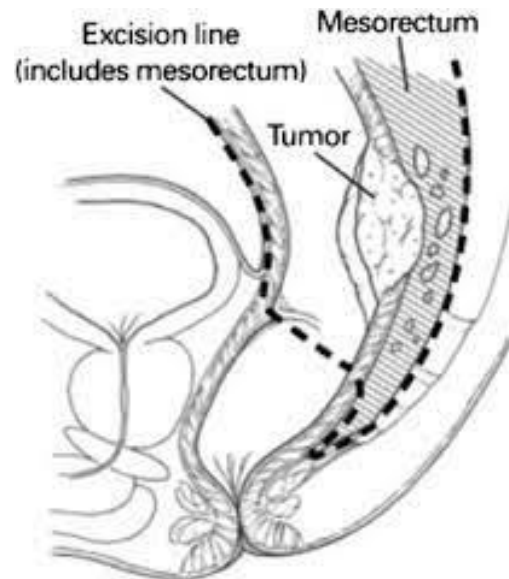
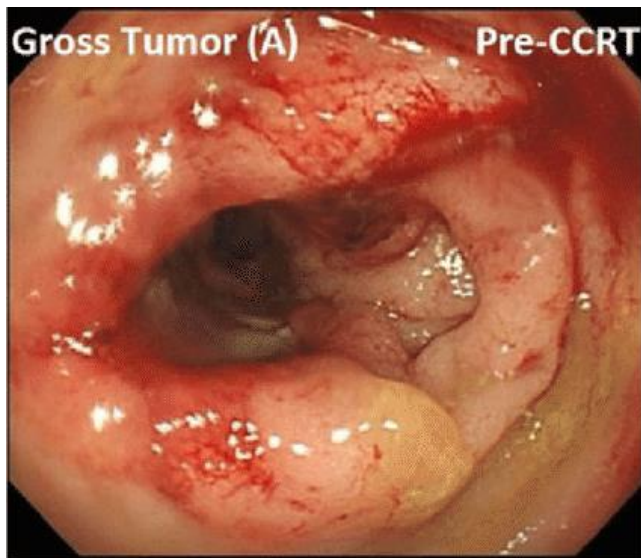
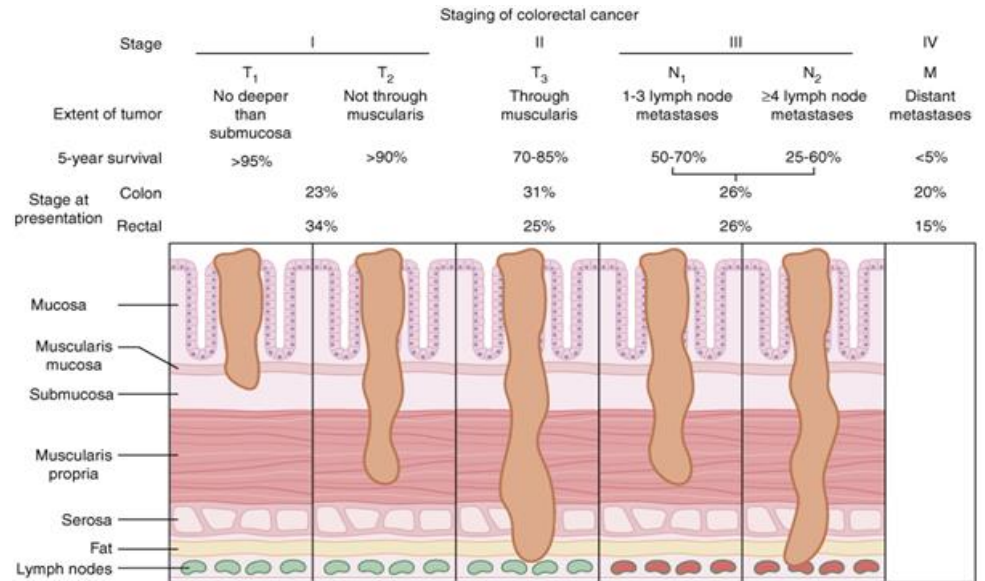
Origin above and below dentate line
(internal and external rectal plexus)



Rectal cancer

Signs and symptoms:

- A change in bowel habits, such as diarrhea, constipation or more-frequent bowel movements
- Dark maroon or bright red blood in stool
- Narrow stool
- A feeling that your bowel doesn't empty completely
- Abdominal pain
- Unexplained weight loss
- Weakness or fatigue



ANAL CANAL - GENERAL

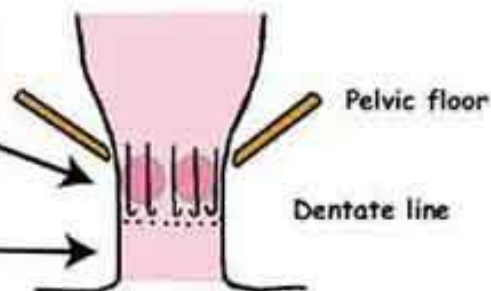
- 4cm long, from pelvic floor (puborectalis) to outside
- Two distinct halves of 2cm separated by dentate (pectinate) line

Upper half (2cm)

- 12 anal columns/valves
- 3 cushions
- Insensitive to touch

Lower half (2cm)

- Skin
- Sensitive to touch



3 spongy mucosal cushions in upper half, level with venous plexuses at 3, 7 & 11 o'clock. Contain arterial & venous blood. Help with continence, air tightness & mucus production. Enlargement leads to haemorrhoids (piles)

UPPER HALF

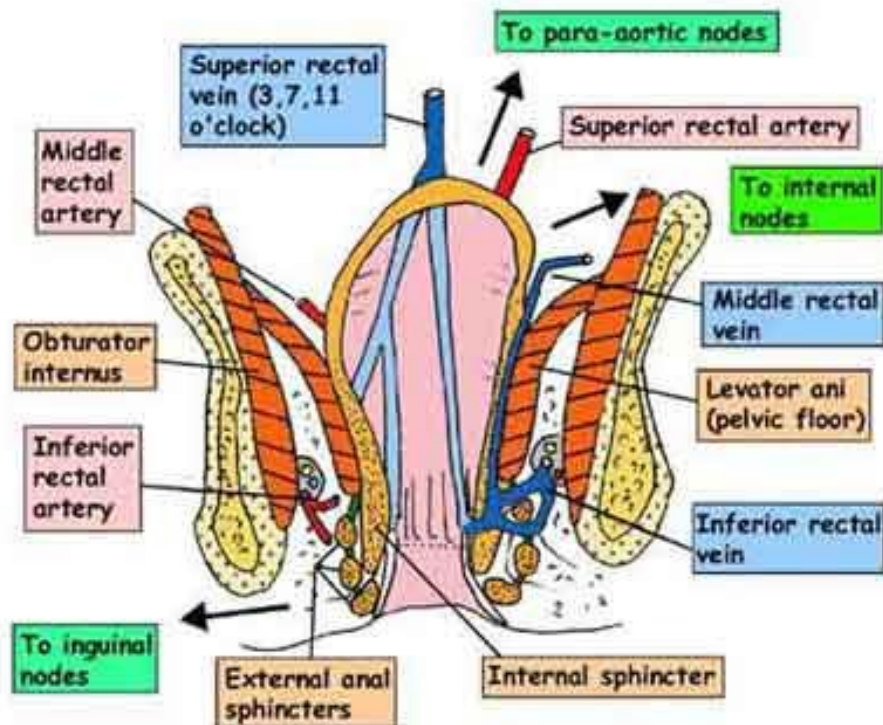
- Endoderm origin
- Columnar mucosa
- Columns, valves & cushions
- Autonomic nerves
- Mainly superior rectal artery
- Portal venous drainage
- Para-aortic lymph nodes
- Adenocarcinoma
- Site of haemorrhoids

LOWER HALF

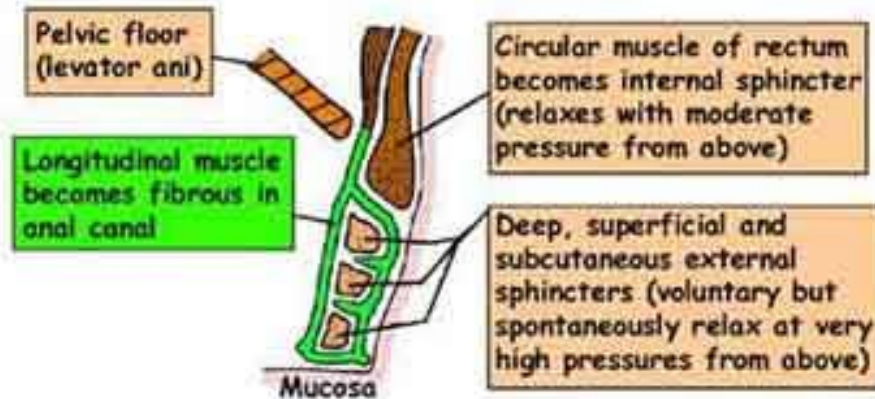
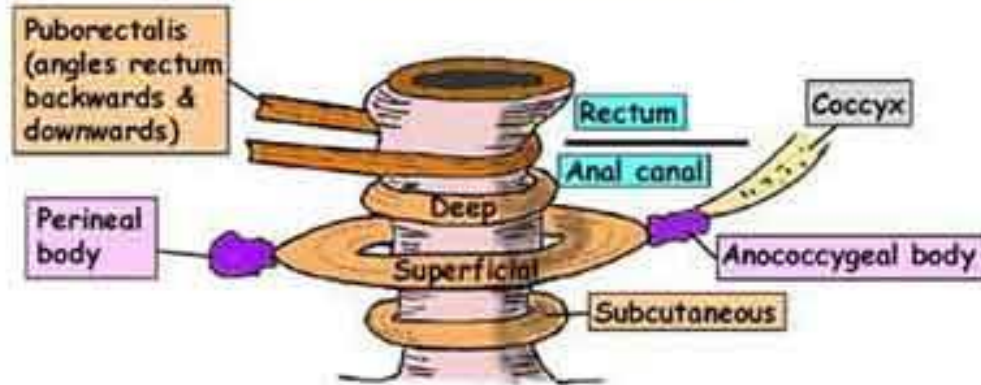
- Ectoderm origin
- Squamous mucosa
- Skin
- Somatic nerves
- Mainly inferior rectal artery
- Systemic venous drainage
- Superficial inguinal nodes
- Squamous carcinoma
- No haemorrhoids

RECTUM - VESSELS/LYMPHATICS

- **Blood supply:** Superior rectal artery from inferior mesenteric
Middle rectal artery from internal iliac. May be small
Inferior rectal artery from internal pudendal
Median sacral may contribute
All arteries supply all layers
- **Venous drainage:** Superior rectal vein to inferior mesenteric which is portal. Middle rectal to internal iliac (systemic)
Inferior rectal to internal pudendal to internal iliac (systemic)
- **Portosystemic anastomosis** In upper anal canal where internal & external venous plexuses meet. Superior rectal vein (portal) meet middle/inferior (systemic)
- **Lymphatics:** Follow deep veins and arteries (black arrows below)



ANAL CANAL - SPHINCTERS



CONTINENCE

Internal sphincter (involuntary)
 External sphincter (voluntary)
 Recto-anal angle (puborectalis)
 Anal cushions & mucosal folds
 Abdominal pressure on upper
 anterior part of lower rectum

NOTE:

Incontinence can be due
 to overflow around
 impacted faeces in
 constipation